

AI Usage Reflection

PMP Solutions

Introduction

Throughout the development of the PMP Solutions AI Fraud Intelligence Platform, our team used Artificial Intelligence (AI) as a collaborative development assistant to improve our understanding, productivity, problem-solving, and the overall quality of our solution. Rather than replacing our own work, AI supported us by accelerating research, explaining technical concepts, debugging code, improving documentation, and helping us make informed technical decisions throughout the Software Development Life Cycle (SDLC).

How AI Supported the Project

AI helped us refine the project scope, compare technologies such as React, Plotly, and Streamlit, understand the PaySim dataset, explain transaction features, guide data cleaning, recommend EDA visualisations, assist with feature engineering, explain machine learning algorithms, model evaluation metrics, SHAP explainability, debug Python code, troubleshoot Google Colab and Jupyter Notebook issues, structure our Streamlit dashboard, organise our GitHub repository, and improve our README, SDLC, executive summary, presentation, and technical documentation.

AI Prompt Usage and Decision-Making

Throughout the project, our team used AI through an iterative process of prompting, evaluating responses, modifying suggestions, and validating outputs. AI served as a collaborative assistant rather than an authoritative source.

Prompts Used

- Explaining the PaySim dataset and transaction features.
- Recommending data cleaning techniques.
- Suggesting EDA visualisations.
- Creating feature engineering strategies.
- Explaining feature selection and data leakage.
- Comparing Logistic Regression, Decision Tree, Random Forest, and XGBoost.
- Explaining Accuracy, Precision, Recall, F1-Score, ROC-AUC and confusion matrices.
- Assisting with SHAP explainability.
- Debugging Python, Google Colab, Jupyter Notebook, Plotly, Streamlit and GitHub.
- Assisting with documentation, SDLC, README, executive summary and presentation preparation.

AI Suggestions Accepted

We accepted AI recommendations that improved data cleaning, feature engineering, visualisations, model comparison, dashboard design, documentation, and debugging after testing and validating them.

AI Suggestions Modified

Many AI suggestions were adapted to suit our dataset, simplify visualisations, improve dashboard usability, refine business recommendations, and align documentation with our team's understanding and project requirements.

AI Suggestions Rejected

Some AI recommendations were not implemented because they introduced unnecessary complexity, duplicated existing visualisations, did not align with the PaySim dataset, or were outside the scope and timeline of the hackathon.

Responsible Use of AI

Every AI-generated explanation, recommendation, code snippet, and document was reviewed, tested, validated, and adapted before inclusion in the final project. The responsibility for analysis, implementation, testing, interpretation, and presentation remained entirely with our team.